

HOW MUCH OF INTERVIEWER VARIANCE IS REALLY NONRESPONSE ERROR VARIANCE?

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Abstract Kish's (1962) classical intra-interviewer correlation (ρ_{int}) provides survey researchers with an estimate of the effect of interviewers on variation in measurements of a survey variable of interest. This correlation is an undesirable product of the data collection process that can arise when answers from respondents interviewed by the same interviewer are more similar to each other than answers from other respondents, decreasing the precision of survey estimates. Estimation of this parameter, however, uses only respondent data. The potential contribution of variance in nonresponse errors between interviewers to the estimation of ρ_{int} has been largely ignored. Responses within interviewers may appear correlated because the interviewers successfully obtain cooperation from different pools of respondents, not because of systematic response deviations. This study takes a first step in filling this gap in the literature on interviewer effects by analyzing a unique survey data set, collected using computer-assisted telephone interviewing (CATI) from a sample of divorce records. This data set, which includes both true values and reported values for respondents and a CATI sample assignment that approximates interpenetrated assignment of subsamples to interviewers, enables the decomposition of interviewer variance

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in means of respondent reports into nonresponse error variance and measurement error variance across interviewers. We show that in cases where there is substantial interviewer variance in reported values, the interviewer variance may arise from nonresponse error variance across interviewers.

Introduction

Survey research organizations conducting interviewer-administered surveys often find that estimates of key population parameters tend to vary across interviewers. Ideally, all interviewers working a random subset of the entire sample obtain a 100 percent response rate, and sampling variance is the only source of variance in measurements between interviewers. Empirical evidence, however, is to the contrary in both telephone and interpenetrated face-to-face surveys (Davis and Scott 1995; Schnell and Kreuter 2005): Respondents interviewed by the same person tend to provide more similar responses for some survey questions than respondents interviewed by different persons (e.g., Groves and Magilavy 1986; Hansen, Hurwitz, and Bershada 1960; O'Muircheartaigh and Campanelli 1998).

One possible source of this between-interviewer variance is correlated deviations of responses from true values within each interviewer (e.g., Biemer and Stokes 1991; Biemer and Trewin 1997; Groves 2004, Chapter 8; Hansen, Hurwitz, and Bershada 1960). Various hypotheses have been proposed in the literature concerning the source of these correlations, including the complexity of the survey question (e.g., Collins and Butcher 1982), interactions between the interviewer and the respondents (e.g., Mangione, Fowler, and Louis 1992), or an inability to disentangle geographic effects from interviewer effects without interpenetrated designs (O'Muircheartaigh and Campanelli 1998). In this article, we provide empirical support for an alternative hypothesis, motivated by consistent findings of between-interviewer variance in response rates (Campanelli and O'Muircheartaigh 1999; Durrant et al. 2010; O'Muircheartaigh and Campanelli 1999; Pickery and Loosveldt 2002; Singer and Frankel 1982). We find that intra-interviewer correlations among respondents on some items arise from variable nonresponse errors across interviewers, rather than correlated response deviations. Surprisingly, to our knowledge, this study is the first to formally examine this hypothesis.

Between-interviewer variance is a well-documented source of instability in survey estimates, decreasing data quality (Groves 2004, 364). The multiplicative "interviewer effect" on the variance of an estimated mean is similar to a design effect due to cluster sampling. This multiplicative effect on the variance is written as $1 + (\bar{m} - 1)\rho_{int}$, where $\bar{m}$ represents the average sample workload for an interviewer and ρ_{int} represents an item-specific intra-interviewer correlation (Kish 1962). When responses to a particular question are more similar for

respondents interviewed by the same interviewer than for respondents interviewed by different interviewers, ρ_{int} can become high, ranging up to 1 (in theory). When responses to a particular question for respondents interviewed by the same interviewer are equivalent to responses obtained from a simple random sample of the total respondent pool, ρ_{int} approaches 0. Low intra-interviewer correlations, however, can substantially affect the precision of an estimated mean. For example, assuming 30 respondents per interviewer, a value of 0.01 for ρ_{int} would result in a 29-percent increase in the variance—or a 13.6 percent increase in the standard error—of an estimated mean, increasing confidence interval width and reducing effective sample sizes.

In practice, empirical estimates of ρ_{int} generally vary from small (0.01) to much larger (0.12 or above). Groves and Magilavy (1986) report that roughly 80 percent of all ρ_{int} estimates fall below 0.02; unfortunately, much of the published literature fails to report how many of these estimates represent *significant* (i.e., non-zero) variability across interviewers. Estimates of ρ_{int} tend to be larger in face-to-face surveys than in telephone surveys (Groves and Magilavy 1986) and are generally on par with or larger than the effects of clusters in area probability sample designs (Davis and Scott 1995; O'Muircheartaigh and Campanelli 1998; Schnell and Kreuter 2005). Many estimates of ρ_{int} are larger than 0.02 for *factual* survey items; estimates of ρ_{int} for attitudinal items tend to be similar to those for factual items (e.g., Groves and Magilavy 1986; Kish 1962; Mangione, Fowler, and Louis 1992). Estimates of ρ_{int} ranging from 0.03 to 0.12 have been reported for factual survey items, including age (Kish 1962, face-to-face), ethnicity (Fellegi 1964, face-to-face), present employment (Groves and Kahn 1979, telephone), existence of health conditions (Mangione, Fowler, and Louis 1992, face-to-face), recent doctor visits (Groves and Magilavy 1986, telephone), and car buying and type of school last attended full-time (Collins and Butcher 1982, face-to-face). Surprisingly, O'Muircheartaigh and Campanelli (1998) report significant intra-interviewer correlations for a series of *self-completion* items in a face-to-face survey.

One possible hypothesis for these unexpected findings is that nonresponse error variance occurs between interviewers. We define *nonresponse error variance* in this context as the variance across interviewers of the nonresponse biases specific to each interviewer's subsample of cases when he or she achieves less than 100 percent response rates. Under this hypothesis, correlations among respondents interviewed by the same person arise because of differential nonresponse errors across interviewers (i.e., interviewers successfully recruit and interview different types of people), not because interviewers introduce correlations in responses to the survey questions (i.e., systematic measurement errors that differ across interviewers). Although suggested previously, especially in the case of telephone surveys (Groves and Fultz 1985, 45; Stokes and Yeh 1988, 358), this hypothesis has not been formally evaluated.

Variation across interviewers in response, contact, and cooperation rates has been clearly documented (Hox and de Leeuw 2002; Link 2006; Morton-Williams 1993; O'Muircheartaigh and Campanelli 1999; Snijkers, Hox, and de Leeuw 1999; Wiggins, Longford, and O'Muircheartaigh 1992). Empirical examinations of the variable effects of interviewers on nonresponse focus on the association between survey participation rates and fixed characteristics of the interviewers (Groves and Couper 1998), their attitudes (Durrant et al. 2010; Hox and de Leeuw 2002), doorstep behaviors (Campanelli, Sturgis, and Purdon 1997; Morton-Williams 1993), and vocal characteristics (Groves et al. 2008; Oksenberg and Cannell 1988). Yet whether interviewers vary in the types of respondents that they recruit has received little research attention, due largely to a lack of information about nonrespondents and the absence of record values for all sample units.

Expanded statistical models for the joint effects of nonresponse and correlated measurement errors on the variance of survey estimates have been posited (Biemer 1980; Groves and Magilavy 1984; Lessler and Kalsbeek 1992; Platek and Gray 1983), although estimation of ρ_{int} in practice typically uses only respondent data. For example, Platek and Gray express an unadjusted estimate of a population total based on respondent reports on a variable x as a function of a participation indicator R_j (where $R_j = 1$ when sample person j participates and $R_j = 0$ when person j is a nonrespondent), the individual's true value on the survey variable Y_j , and their individual response deviation, e_j : $\hat{x} = \frac{N}{n} \sum_{j=1}^n R_j (Y_j + e_j)$. Platek and Gray show that in the absence of any type of weighting or imputation method to compensate for nonresponse (which they refer to as the "zero substitution" method), the variance of this unadjusted linear statistic can be decomposed into five terms: sampling variance, uncorrelated measurement error variance (i.e., simple response variance), correlated measurement error variance, uncorrelated nonresponse error variance, and correlated nonresponse error variance.

We present their general derivation of this variance below (Platek and Gray 1983, 287, 316), using the notation p_{ij} for the response propensity of person j assigned to interviewer i , σ_{ij} for the standard deviation of the response deviations for person j assigned to interviewer i , B_{ij} for a person's average response deviation, $r_{2ijj'}$ for the correlation of the response deviations for two sampled persons j and j' assigned to interviewer i , and $r_{1ijj'}$ for the correlation of the participation indicators R for persons j and j' assigned to interviewer i . To emphasize the nesting of sample persons within interviewers, we use A for the total number of interviewers in the population, and M for the fixed subsample size assigned to each interviewer:

$$\begin{aligned}
 V(\hat{x}) = & \text{Sampling Variance} \\
 & + \text{Simple Measurement Error (ME) Variance} \\
 & + \text{Correlated Measurement Error (ME) Variance} \quad (\text{Equation PG}) \\
 & + \text{Simple Nonresponse Error (NE) Variance} \\
 & + \text{Correlated Nonresponse Error (NE) Variance}
 \end{aligned}$$

where

$$\begin{aligned} \text{Sampling Variance} &= \left(\frac{N}{n}-1\right) \sum_{i=1}^A \sum_{j=1}^M (p_{ij}(Y_{ij} + B_{ij}))^2 \\ &+ \left(\frac{N(n-1)}{n(N-1)}-1\right) \sum_{i=1}^A \sum_{j \neq j'}^M (p_{ij}(Y_{ij} + B_{ij}))(p_{ij'}(Y_{ij'} + B_{ij'})); \end{aligned}$$

$$\text{Simple ME Variance} = \frac{N}{n} \sum_{i=1}^A \sum_{j=1}^M [p_{ij}^2 + p_{ij}(1-p_{ij})] \sigma_{ij}^2;$$

$$\begin{aligned} \text{Correlated ME Variance} &= \frac{N(n-1)}{n(N-1)} \sum_{i=1}^A \sum_{j \neq j'}^M r_{2ijj'} \sigma_{ij} \sigma_{ij'} \\ &\quad \left[p_{ij} p_{ij'} + r_{1ijj'} \sqrt{p_{ij}(1-p_{ij}) p_{ij'}(1-p_{ij'})} \right]; \end{aligned}$$

$$\text{Simple NE Variance} = \frac{N}{n} \sum_{i=1}^A \sum_{j=1}^M p_{ij}(1-p_{ij})(Y_{ij} + B_{ij})^2; \text{ and}$$

$$\begin{aligned} \text{Correlated NE Variance} &= \frac{N(n-1)}{n(N-1)} \sum_{i=1}^A \\ &\quad \sum_{j \neq j'}^M r_{1ijj'} \sqrt{p_{ij}(1-p_{ij}) p_{ij'}(1-p_{ij'})} (Y_{ij} + B_{ij})(Y_{ij'} + B_{ij'}). \end{aligned}$$

This daunting expression shows that under the normal situation of less than 100 percent response rates, the correlated measurement error variance component of a linear statistic contains correlations arising from (1) correlations among response deviations between two respondents; and (2) correlations among the participation indicators between two different units. Correlated nonresponse error variance in the last term of the equation arises from correlations among indicators for survey participation, R_{ij} and $R_{ij'}$, between two different units assigned to the same interviewer.

Applying these models to the estimation of nonresponse and measurement error variance due to the interviewer is complicated. Platek and Gray argue that interviewers are one potential source for these correlations, but interviewers are not explicitly accounted for in the model. Further, few of these models have been applied to nonsimulated survey data or non-imputed data. In fact, Platek and Gray illustrate their formulas by assuming values for many of the terms rather than estimating them, including the correlations among response deviations and among participation indicators. Our review of the literature found that none of the past work in this area has developed separate estimators for the correlated components of measurement error variance and

nonresponse error variance for unadjusted estimators (although some work has been done using imputed data). Further, no work has extended these results to explicitly account for the role of interviewers in introducing correlated measurement errors and correlated response indicators, despite several examples of interviewers introducing correlations of both forms in the literature.

Despite these complications, the models that have been proposed for the joint effects of correlated nonresponse and measurement errors on survey estimates are particularly useful in that they elucidate characteristics of the data that are needed for estimating these components. First, each interviewer employed for the survey should be assigned a completely random interpenetrated subsample of the full sample selected for the survey (Mahalanobis 1946). Second, important survey variables must be available for both respondents and nonrespondents (perhaps on the original sampling frame) and measured without error. Third, these important survey variables should be asked in identical form in the survey questionnaire to minimize any systematic response biases. We consider a survey data set having these unique properties in this study.

In this article, we present an initial descriptive examination of the variance in these two critical error sources among interviewers, in hopes of motivating future analytical development. We analyze a unique survey data set that enables estimation of correlated interviewer variance due to measurement error and nonresponse error. The data arise from a survey administered to a sample drawn from a frame of records containing items that are also asked in the survey. Sample cases are assigned to interviewers based on an interpenetrated sample design. Knowledge of true values on survey variables for the respondents and nonrespondents permits calculation of nonresponse error variance due to interviewers. We use these data to show that respondent means may vary between interviewers not because of correlated response deviations within interviewers, but because interviewers successfully obtain participation from different pools of respondents.

Data and Methods

Data: The survey data used here were collected by the University of Wisconsin–Madison as part of the Wisconsin Divorce Study (WDS). The WDS selected a simple random sample of 733 divorce certificates from a frame of certificates in 1989 and 1993 from four counties in Wisconsin. The sampled divorce certificates included official information that was also collected in the survey. The present study considers data collected from 355 respondents (AAPOR RR1 = 71.0 percent) by 31 trained interviewers¹ using computer-assisted telephone interviewing (CATI), meaning that interviewer effects will likely be attenuated in this analysis relative to a face-to-face survey (Groves and Magilavy 1986).

1. Additional interviewer-level information, such as interviewing experience, is not available.

The divorce date on each certificate was recorded by an official body; other information on the frame (e.g., marriage date, birth dates) was reported by at least one member of the divorcing couple (not indicated in the records) and therefore could be more susceptible to measurement error. For purposes of this study, we assume that the frame data are essentially error-free. The questionnaire itself collected measures on six variables that will be analyzed in this study with respect to the frame records: length of marriage in months (divorce date minus marriage date), time since divorce in months (interview date minus divorce date), time since marriage in months (interview date minus marriage date), number of marriages up to and including the divorce, age at marriage (marriage date minus birth date), and age at divorce (divorce date minus birth date). We note that five of these six variables are functions of dates collected in the survey, which is common in survey practice. The WDS question wording for these variables was “How many times have you been married?”; “In what month and year did your marriage begin?”; “In what month and year did you get divorced?”; and “What is your date of birth?”

The call records in the WDS were kept using paper-and-pencil cover sheets for each sample case, and the call-record data were entered using double entry. Illegible interviewer handwriting resulted in limited amounts of item-missing data in the call records. The interviewer also recorded his or her initials at each call. In some instances, subjective decisions about the interviewer identification codes associated with each call had to be made (e.g., an interviewer recorded two initials for one call and then three initials for another call). Interviewer initials in the electronic data file that were deemed by the authors to correspond to the same person were combined. All analyses in this study were performed before and after this combining operation; no substantive differences in the results were found. Important variables from the call records for the present study include the initials of the interviewer making the call attempt, the number of call attempts to a particular sample case, the time of the call attempt, the day of the week of the call attempt, and the call outcome.

Assigning sampled cases to interviewers: Ideally, for purposes of estimating nonresponse error variance among interviewers, each interviewer in the WDS would have had a fixed random subsample of cases to work and never made call attempts to cases being worked by other interviewers (e.g., Singer and Frankel 1982). In practice, however, any CATI survey involves frequent changes in the interviewer who works the sampled case. As a result, how to assign cases to interviewers for estimating components of variance due to nonresponse error is a nontrivial decision.

In this study, we assumed that interviewers working a particular shift (e.g., weekdays, 9–5 p.m.) were assigned random subsamples of all sampled persons called during that shift. This assumption of interpenetration within shifts was empirically tested for each survey variable of interest (discussed below). Assigning interviewers to discrete shifts is difficult in CATI surveys because

the work of interviewers often extends across multiple calling shifts. We conducted all of our analyses conditional on a calling shift; that is, we estimated variance between interviewers within a shift. Because persons with different characteristics are likely to be contacted at different times of the day (e.g., employed persons are more likely to be contacted during evening hours; Groves and Couper 1998), not accounting for the shift of work could confound non-response error with true differences in sampled persons across shifts.

Assigning shifts: We included only telephone calls with a recorded interviewer making the call and a time and day of call recorded in the data set, leaving 5,267 calls for analysis (94.9 percent of all 5,548 recorded calls). The time and day of each call were combined into a single shift variable denoting three time slots: weekday day, or Monday through Friday before 5 p.m. (Shift 1); weekday evening after 5 p.m. (Shift 2); and weekend calls on Saturday and Sunday (Shift 3). Stokes and Yeh (1988) also used this collapsing of shifts to analyze interviewer variance in a telephone survey.² Overall, 44.1 percent of the calls made in the WDS were made during the weekday evening shift, 29.1 percent during the weekend shift, and 26.8 percent during the weekday day shift. About 90 percent of the interviewers made at least one call during each of the three shifts. Only seven of the 31 interviewers made at least 90 percent of their calls during one of the three shifts, indicating shift crossing similar to that found by Stokes and Yeh (1988, 368).

Assigning nonrespondents to interviewers: Multiple interviewers worked cases in multiple shifts, leading to difficulty in assigning a single interviewer and shift to each case for analysis. For all 355 interviewed cases, the interviewer who conducted the interview was assigned to each respondent, as was the corresponding shift during which the interview was conducted. For nonresponding cases, decisions about how to assign interviewers and shifts were more complicated.

Sample cases that had no call attempts with an interviewer ID recorded in the call records were deleted from the data set. This resulted in the deletion of 54 sample cases with call records but item-missing data on interviewer IDs, while 679 sample cases were retained (of which 355 responded). The remaining 324 sample cases consisted of both refusals and noncontacts. If a sample case was contacted at some point but refused to cooperate, it was assigned to the first interviewer receiving a refusal from that case. If there were no prior explicit refusals for a contacted case, the nonrespondent case was assigned to the last interviewer making contact. The WDS documentation states that prior refusals were assigned to interviewers trained in refusal conversion, but no particular guidelines were followed for cases that had not been previously contacted

2. The present study also considered three alternative shifts: Monday through Friday before 5 p.m., Sunday through Thursday after 5 p.m., and Friday or Saturday after 5 p.m./weekends before 5 p.m. The results from this analysis were qualitatively the same, so we present results from only the first shift assignment.

Table 1. Descriptive Statistics for Interviewer Workloads and Response Rates, by Shift

	Shift 1: Weekday day	Shift 2: Weekday evening	Shift 3: Weekend	Total
# Cases Attempted	163	345	169	677**
# Respondents	67	187	101	355
Response Rate (%)	41.1%	54.2%	59.8%	52.4%
# of Interviewers	20	24	19	30
Mean Interviewer Workload (SD)	8.2 (12.0)	14.4 (15.0)	8.9 (8.9)	10.7 (12.6)

NOTE.—SD = Standard Deviation.

** Two (2) cases did not have a final shift recorded in the call records.

(Mitchell 2004). Therefore, if calls were made to a sample case but contact was never established, the nonrespondent was assigned to the last interviewer making a call, assumed to be randomly selected from all interviewers working a shift.³

Increasing response rates of 41.1 percent, 54.2 percent, and 59.8 percent were found across the three shifts under the nonresponse assignment method described above (table 1). Cooperation rates across the three shifts among contacted cases were fairly stable (75.3 percent, 73.1 percent, and 75.4 percent, respectively). Consistent with previous work examining optimal calling periods (Brick et al. 2007; Groves and Couper 1998; Hu et al. 2009; Weeks, Kulka, and Pierson 1987), this finding suggests that interviewers had a more difficult time making contact during the week, especially before 5:00 p.m.

The number of interviewers who worked each shift ranged from 19 to 24 (table 1). These numbers limit the power of analyses to detect statistically significant variation between interviewers. Substantial variability in assigned workloads (respondents and nonrespondents) across interviewers within a shift is also evident. Mean workloads range from 8.2 sampled cases (Shift 1) to 14.4 sampled cases (Shift 2), and workload standard deviations range from 8.9 (Shift 3) to 15.0 (Shift 2).

Analytic Approach: The traditional interviewer variance model for a survey variable Y in interviewer-administered surveys with 100 percent response rates

3. Assignment of noncontacts to the last interviewer attempting a call attempt may be considered arbitrary. An alternative method of assigning noncontacts was also considered. In this method, an interviewer making at least one call to a case that had not been contacted was selected at random from all of the interviewers making calls to this case and assigned to it. The primary findings in this study did not vary when using this alternative method of assigning noncontacts to interviewers, suggesting that interviewers making the final calls to noncontacted cases were randomly selected from all interviewers. Additional methods of assigning noncontacts were not considered.

expresses an individual response as $x_{ij} = \bar{Y} + b_i + e_{ij}$, where x_{ij} is respondent j 's report collected by interviewer i , $\bar{Y}$ is the population mean of the true values, b_i is the deviation in the respondent's report due to interviewer i , and e_{ij} is a random measurement error term. The variance of a sample mean can then be written (approximately) in terms of the sampling variance, the variance due to interviewers, and random error variance (Biemer and Stokes 1991; U.S. Bureau of the Census 1985, 228; Kish 1962):

$$\text{Var}(\bar{x}) \approx \frac{S_Y^2}{n} [1 + (\bar{m} - 1)\rho_Y] = \frac{\sigma_Y^2 + \sigma_b^2 + \sigma_e^2}{n} [1 + (\bar{m} - 1)\rho_Y]. \quad (1)$$

In equation (1), n is the sample size, σ_Y^2 represents the element variance of the true values of Y , σ_b^2 is the correlated component of variance due to interviewers, σ_e^2 is random error variance, $\bar{m}$ represents the average interviewer workload, and ρ_Y represents the intra-interviewer correlation of variable Y , with $\rho_Y = \frac{\sigma_b^2}{\sigma_Y^2 + \sigma_b^2 + \sigma_e^2}$. Thus, estimates with larger correlated components of variance in responses due to interviewers (σ_b^2) have larger intra-interviewer correlations (ρ_Y). Estimators of ρ_Y have been proposed in the literature based on analysis of variance methods (U.S. Bureau of the Census 1985; Kish 1962) and multilevel modeling methods (e.g., O'Muircheartaigh and Campanelli 1998). These estimators assume 100 percent response rates and focus exclusively on measurement error.

Our analytic approach considered a descriptive method for decomposing interviewer variance in respondent reports into estimates of nonresponse error variance and measurement error variance. We then explored the ratio of nonresponse error variance to total interviewer variance. Let y_{ij} be the true value of survey variable Y for sample unit j assigned to interviewer i , and let x_{ij} be the reported value for Y for that sample unit if the unit responds to the survey. Assuming interpenetrated assignment of subsamples to interviewers, the expectation of the mean of respondent reports $\bar{x}_i$ for interviewer i is

$$E(\bar{x}_i|i) = \bar{Y} + \text{Bias}_{NR,i} + \text{Bias}_{ME,i} = \bar{Y} + (\bar{y}_{R,i} - \bar{y}_i) + (\bar{x}_i - \bar{y}_{R,i}). \quad (2)$$

That is, the expected value of the respondent mean for interviewer i is the sum of (1) the population mean of the true values, $\bar{Y}$; (2) the difference between the mean of the true values for all *respondents* interviewed by interviewer i , $\bar{y}_{R,i}$, and the mean of the true values for all *units* assigned to interviewer i , $\bar{y}_i$ (nonresponse bias); and (3) the difference between the mean of the *reported* values for all respondents interviewed by interviewer i , $\bar{x}_i$, and $\bar{y}_{R,i}$ (measurement error bias).

Assuming interpenetration and negligible covariance between the two bias terms,⁴ the variance of the expectation in (2) is defined by the sum of two

4. We computed the correlations (r) of these two bias sources for interviewers within each shift. The correlations ranged from -0.05 to 0.32 . None were significant ($p < 0.05$), providing empirical support for the assumption.

variance components: $\text{Var}(\text{Bias}_{NR,i})$ and $\text{Var}(\text{Bias}_{ME,i})$. We sought unbiased estimates of these variance components and the ratio of the nonresponse error variance (the first component) to the total interviewer-related variance (the sum of the two components). Estimation of these variance components using closed-form estimators is possible for very simple design and response scenarios (e.g., equal assignment sizes and response rates across interviewers) typically not experienced in practice. Given the unequal assignment sizes and respondent counts for each interviewer in the WDS, we estimated these components for each variable within each calling shift using three distinct steps.

Estimation Step 1. First, we estimated the variance among interviewers in the means of the true values for all sample cases assigned to each interviewer. Assuming interpenetrated assignment of cases to interviewers, this variance component should be negligible, as all interviewers should have a full sample mean of true values equal, on average, to the population mean. We estimated this component using a one-way random effects model, assuming that the interviewers were a random subsample from a larger hypothetical population of interviewers:

$$y_{ij} = \bar{Y} + b_i + e_{ij}. \quad (3)$$

In this notation, y_{ij} is the true value of variable Y for sample unit j assigned to interviewer i , b_i is the random deviation of interviewer i 's assignment mean from the population mean of the true values, $\bar{Y}$, and e_{ij} is a normally distributed random error with mean 0 and constant variance (the element variance within each assignment). We estimated the variance of the b_i , or $\text{Var}(b_i) = \sigma_{\text{int,full}}^2$, using restricted maximum likelihood (REML) estimation to obtain an unbiased estimate of this variance component given unequal interviewer workloads (Paterson and Thompson 1971).

Significance tests for variance components are more complicated than those for means and proportions. A large body of statistical research has been dedicated to appropriate methods for testing the significance of variance components in models including random effects (Zhang and Lin 2008). We tested a null hypothesis that the interviewer variance component is equal to zero, $H_0 : \sigma_{\text{int,full}}^2 = 0$, versus the alternative that assignment means vary across interviewers, $H_A : \sigma_{\text{int,full}}^2 > 0$. One test of this hypothesis is the Wald test, where the REML estimate of the variance component is divided by its asymptotic standard error.⁵ Although intuitively simple, this test behaves poorly due to the complicated distribution of the test statistic under the null hypothesis, among

5. Bates (2009), the developer of the nlme and lme4 packages for fitting mixed-effects models in R, advises against reporting standard errors for estimated variance components when the distributions of the estimators are not symmetric. This applies for REML estimators of variance components. We follow this recommendation.

other reasons (Berkhof and Snijders 2001). A likelihood ratio test⁶ for the variance component has been shown to have better statistical properties (Self and Liang 1987; Stram and Lee 1994), yet recent simulation studies have demonstrated that the likelihood ratio test statistic has a more complicated distribution in many practical situations than previously thought (e.g., Crainiceanu 2008; Greven et al. 2008). As such, we used the exact null distribution of the likelihood ratio test statistic under more general conditions (including small samples, as in this study) (Crainiceanu and Ruppert 2004). We performed finite-sample likelihood ratio tests of the null hypothesis for the models, and reported p -values for the observed restricted likelihood ratio test (RLRT) statistics under the simulated null distributions. All analyses were conducted using the statistical package R (R code available upon request).

Estimation Step 2. Second, when assignments were interpenetrated, each interviewer had the same $\bar{y}_i$ in expectation, and the variance of the nonresponse biases simplifies to $\text{Var}(\text{Bias}_{NR,i}) = \text{Var}(\bar{y}_{R,i} - \bar{y}_i) = \text{Var}(\bar{y}_{R,i})$. We thus estimated the nonresponse error variance component by estimating the variance across interviewers in the means of the true values for *respondents*. We again used a one-way random effects model for the true values of *respondents* to the survey request, $y_{R,ij}$:

$$y_{R,ij} = \bar{Y}_R + b'_i + e'_{ij}. \quad (4)$$

Here, b'_i captures the random deviation of each interviewer's mean for their recruited respondents' true values from the expected value of the mean of the true values for respondents over all possible sample assignments to interviewers ($\bar{Y}_R$). We note that in the absence of overall nonresponse bias, $\bar{Y}_R$ will be equal to the population mean of the true values, $\bar{Y}$. We estimated the variance of these random effects, $\text{Var}(b'_i) = \sigma^2_{\text{int,resp}}$, using REML to obtain an unbiased estimate of $\text{Var}(\text{Bias}_{NR,i})$. We tested this component of variance against zero using the likelihood ratio tests described above.

Estimation Step 3. Third, under an assumption of interpenetrated assignment of subsamples to interviewers, equation (2) can be rewritten as

$$\begin{aligned} E(\bar{x}_i|i) &= \bar{Y} + \text{Bias}_{NR,i} + \text{Bias}_{ME,i} = \bar{Y} + (\bar{y}_{R,i} - \bar{Y}) + (\bar{x}_i - \bar{y}_{R,i}). \\ &= \bar{x}_i \end{aligned} \quad (5)$$

Using the sample mean of the respondent reports for interviewer i as an estimate of this expectation, we then computed an unbiased estimate of the variance in the means of the reported values across interviewers (equation (5)) using a one-way random effects model and REML estimation:

6. Models including and excluding the random interviewer effects are estimated. The test compares the positive difference in the -2 REML log-likelihood values for the two models (the restricted likelihood ratio test statistic, or RLRT statistic, under the null hypothesis) to a mixture of χ^2 distributions.

$$x_{ij} = \bar{X}_R + b_i'' + e_{ij}''. \tag{6}$$

This is the interviewer variance model that is often estimated in practice using respondent data only. The variance of the random interviewer deviations (b_i'') around the expected value of the mean of the respondent reports over all possible sample assignments to interviewers ($\bar{X}_R$), or $Var(b_i'') = \sigma_{int,resp,obs}^2$, captures both measurement error variance and nonresponse error variance introduced by the interviewers, and is thus the “total variance” due to interviewers. We tested this variance component for significance using appropriate finite-sample likelihood ratio tests. We then subtracted the estimate of the nonresponse error variance from Estimation Step 2 to get an estimate of the measurement error variance across interviewers. The estimated proportion of variance introduced by interviewers due to nonresponse error variance was then computed as

$$\frac{\hat{\sigma}_{int,resp}^2 - \hat{\sigma}_{int,full}^2}{\hat{\sigma}_{int,resp,obs}^2 - \hat{\sigma}_{int,full}^2}. \tag{7}$$

Evidence of successful interpenetration from Estimation Step 1 implies that $\sigma_{int,full}^2 = 0$ (i.e., interviewer-level means of true values for their full assignments do not vary). We subtracted estimates of $\sigma_{int,full}^2$ from the numerator and denominator to remove components of variance that were not due to the interviewer from this calculation.

Results

For each of the three WDS calling shifts, [table 2](#) displays the sample size, number of interviewers, and number of responding cases. The number of responding

Table 2. Total Counts of Interviewers, Sampled Cases, and Item Respondents, by Shift

	Shift 1: Weekday day	Shift 2: Weekday evening	Shift 3: Weekend
Number of Interviewers			
Total	20	24	19
Responding Cases Only	17	23	17
Number of Sampled Persons			
Frame Total	163	345	169
Item Respondents			
Minimum	53	161	87
Maximum	67	187	101

cases varies slightly over items due to differential item nonresponse. We have limited power to detect variance among interviewers within each shift, given the small counts of interviewers and the small number of cases assigned to each interviewer.

Table 3 presents estimates of the three variance components of interest and the estimated proportion of the interviewer variance due to nonresponse error variance across interviewers for each WDS variable in each shift. We first consider the assumption of interpenetrated assignment of subsamples of active sample cases to interviewers within each calling shift. Estimates of the variance in means of true values for cases assigned to each interviewer are provided in column 3. We find little evidence against assumptions of interpenetration within each shift, as very little variance occurs among interviewers in the means of the assigned subsamples. Although there is low power to detect significant differences, the magnitude of the estimated variance term $\sigma_{\text{int,full}}^2$ is very small for almost all of the variable/shift combinations. In three cases (age at divorce in Shift 1, months since marriage in Shift 2, and months since divorce in Shift 2), there is evidence of marginal variance among interviewers in assignment means, suggesting that the assumption of interpenetrated assignment did not appear to hold for all variables in all shifts.

Next, we examine interviewer variance due to nonresponse error and measurement error. We find two variable/shift combinations presenting evidence of significant interviewer variance in the means of respondent reports, despite evidence of successful interpenetration: age at marriage in Shift 2 (weekday evening), and age at divorce in Shift 2. For age at marriage in Shift 2, we see negligible variance among interviewers in the means of true values for responding cases, suggesting negligible nonresponse error variance. There is significant ($p < 0.01$) variance among interviewers in the means of *reported* values for respondents, which suggests that the majority of the interviewer variance arises from measurement error variance. This is consistent with the long-standing hypothesis that interviewer variance arises due to correlations among response deviations for respondents interviewed by the same interviewer.

For age at divorce in Shift 2, we see evidence of significant ($p = 0.05$) variance between interviewers in the means of true values for respondents alone. Thus, different interviewers successfully recruited respondents having different ages at divorce. The estimate of the interviewer variance in means of reported values for respondents was only slightly higher and remained significant ($p = 0.03$), suggesting little added variance from measurement error variance among the interviewers. Overall, an estimated 80.6 percent of the variance added by interviewers in this case was due to nonresponse error variance.

In several cases, the estimates of nonresponse error variance or the total interviewer variance are extremely close to zero, preventing reliable estimation of the proportion of interviewer variance due to nonresponse error variance. For the remaining eight variable/shift combinations where the estimated proportion of interviewer variance due to nonresponse error variance could be computed,

Table 3. Estimates of Interviewer Variance Components for Each WDS variable in Each WDS Shift, and Estimated Proportions of Interviewer Variance Due to Nonresponse Error Variance across Interviewers

WDS Variable	Shift	Est'd Variance in True Values of Subsamples:	RLRT Statistic	Est'd Variance in True Values of Resp.:	RLRT Statistic	Est'd Variance in Reported Values of Resp.:	RLRT Statistic	Est'd Proportion of Interviewer Variance Due to NR Error Variance:
		$\hat{\sigma}_{int,full}^2$		$\hat{\sigma}_{int,resp}^2$		$\hat{\sigma}_{int,resp,obs}^2$		$\frac{\frac{\hat{\sigma}_{int,resp}^2}{2} - \frac{\hat{\sigma}_{int,full}^2}{2}}{\frac{\hat{\sigma}_{int,resp,obs}^2}{2} - \frac{\hat{\sigma}_{int,full}^2}{2}}$
Age at Marriage	1	0.55	0.38	1.73	0.31	2.58	0.55	0.583
	2	0.55	0.52	0.00	0.00	3.66	7.26***	0.000
	3	0.00	0.00	0.00	0.00	0.00	0.00	N/A
Age at Divorce	1	4.44	2.06*	1.79	0.02	1.38	0.01	N/A
	2	0.74	0.46	3.57	2.27**	4.25	3.18**	0.806
	3	0.00	0.00	3.05	0.30	4.07	0.50	0.749
Months Since Marriage	1	0.00	0.00	0.00	0.00	570.50	0.28	0.000
	2	234.05	1.74*	287.07	1.09	277.38	0.50	N/A
	3	0.00	0.00	0.00	0.00	0.00	0.00	N/A
Months Since Divorce	1	0.00	0.00	0.00	0.00	0.00	0.00	N/A
	2	12.07	2.36**	17.51	0.65	35.18	0.73	0.235
	3	0.00	0.00	27.82	0.48	33.09	0.39	0.841
Length of Marriage	1	0.00	0.00	0.00	0.00	0.00	0.00	N/A
	2	134.76	0.77	168.00	0.43	204.17	0.51	0.479
	3	38.28	0.02	0.00	0.00	0.00	0.00	N/A

Continued

Table 3. Continued

WDS Variable	Shift	Est'd Variance in True Values of Subsamples: $\hat{\sigma}_{int,full}^2$	RLRT Statistic	Est'd Variance in True Values of Resp.: $\hat{\sigma}_{int,resp}^2$	RLRT Statistic	Est'd Variance in Reported Values of Resp.: $\hat{\sigma}_{int,resp,obs}^2$	RLRT Statistic	Est'd Proportion of Interviewer Variance Due to NR Error Variance: $\frac{\hat{\sigma}_{int,resp}^2 - \hat{\sigma}_{int,full}^2}{\hat{\sigma}_{int,resp,obs}^2 - \hat{\sigma}_{int,full}^2}$
Number of Marriages	1	0.00	0.00	0.02	1.48*	0.00	0.00	N/A
	2	<0.01	0.98	<0.001	0.00	0.007	1.06	0.000
	3	<0.01	0.02	<0.01	0.34	0.01	0.44	0.749

* $p \leq 0.10$; ** $p \leq 0.05$; *** $p \leq 0.01$.

NOTE.— The p -value for the restricted likelihood ratio test (RLRT) statistic is based on the finite sample distribution of the RLRT statistic under the null hypothesis that the variance component is equal to 0 (Crainiceanu and Ruppert 2004) and is computed using 10,000 simulations from the null distribution.

N/A: The estimated proportion of interviewer variance due to nonresponse error variance among interviewers could not be computed, for one of two reasons: (1) the interviewer variance components were consistently estimated to be extremely small (very close to 0); or (2) the estimated nonresponse error variance was *greater* than the total interviewer variance based on respondent data. This may have occurred if respondent reports attenuated the existing nonresponse error variance, rather than adding to it, or if a non-zero correlation between the bias components existed (ranging from -0.05 to 0.32 in this study). Additional research into these unusual findings is needed.

four of the estimated proportions were greater than 50 percent, which suggests that more variance was being introduced by nonresponse error than by measurement error. However, the tests of significance for the total interviewer variance in these eight cases all fail to reject the null hypothesis of no interviewer variance in the means of respondent reports. This could be a function of the limited power that we had to detect interviewer variance in these shifts. We include these results to motivate studies with more power to examine the mix of nonresponse error variance and measurement error variance in total interviewer variance.

Figure 1 compares the distributions of interviewer-specific nonresponse errors and measurement errors for age of marriage and age of divorce in Shift 2. In each shift, the nonresponse error is computed for each interviewer i as $(\bar{y}_{R,i} - \bar{y}_i)$, and the measurement error for each interviewer is computed as the mean response deviation. The errors for the interviewers are weighted by their assigned subsample sizes in the box plot.

The nonresponse error variance among interviewers is much larger than the measurement error variance for the age at divorce variable (in Shift 2). For the age at marriage variable in Shift 2, the increased measurement error variance appears to be driven partially by a single unusual interviewer who collected extreme values on the age at marriage measures. Even after excluding this

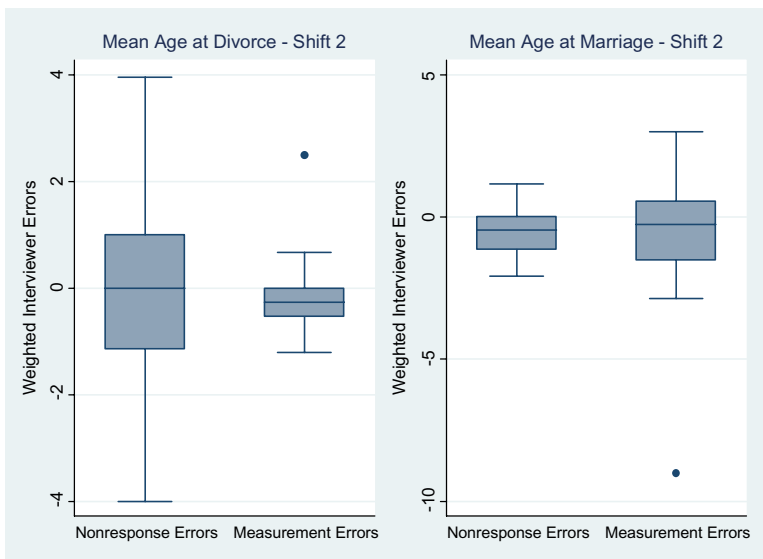


Figure 1. Box Plots Showing Nonresponse Error Variance and Measurement Error Variance among Interviewers for the Two Variable/Shift Combinations with Evidence of Significant Interviewer Variance Based on Respondent Data, with Interviewers Weighted by Their Assigned Subsample Sizes in Each Shift.

extreme interviewer, the variance of the means of the respondent reports across interviewers was still significant ($p < 0.01$), suggesting that measurement error variance was the primary source of the interviewer variance for this variable in this shift.

Given these results, we revisit the variance of an estimated sample total presented by Platek and Gray (equation *PG*). We note that in the case of no measurement error (i.e., $B_{ij} = 0$ for all respondents), the correlated nonresponse error variance term can be re-expressed as the covariance of true values within interviewers for different respondents: $\frac{N(n-1)}{n(N-1)} \sum_{i=1}^A \sum_{j \neq j'}^M \text{cov}(R_{ij}Y_{ij}, R_{ij'}Y_{ij'})$. In a one-way random effects model, the variance of the random interviewer effects is equivalent to the marginal covariance of two values within the same interviewer (see West, Welch, and Galecki 2007, Appendix B); we estimate this component in table 3 ($\hat{\sigma}_{\text{int,resp}}^2$). In the absence of measurement error, the contribution of interviewers to the total variance is thus defined by $\hat{\sigma}_{\text{int,resp}}^2$. Allowing for measurement error in the respondent reports, $r_{2ijj'}\sigma_{ij}\sigma_{ij'}$ in equation *PG* is the between-interviewer variance in the mean response deviations. We can re-express this term as the covariance of response deviations from two respondents interviewed by the same interviewer. To approximate this covariance, we can compute $\hat{\sigma}_{\text{int,resp,obs}}^2 - \hat{\sigma}_{\text{int,resp}}^2$ in table 3, assuming negligible covariance between bias terms.

We also used the WDS data to estimate the intra-interviewer correlations in response indicators and response deviations for these two variables measured in Shift 2.⁷ For age at divorce, these estimates were 0.175 and 0.041, respectively, while for age at marriage, these estimates were 0.176 and 0.043. Platek and Gray, citing an absence of empirical information in the literature on these correlations of response indicators and response deviations between different sample persons, used 0.05 as an average correlation when illustrating how to calculate the variance (equation *PG*) under different design scenarios. In those examples, Platek and Gray (1983, 301) found that the proportion of total variance due to nonsampling variance ranged from 0.31 to 0.44, and stated that when the two average correlations were fixed to 0.05, “the bulk of the nonsampling variances are due to the errors among individual units rather than pairs of units. . .” (304). As reported here and in other studies of interviewer variance in response rates (e.g., O’Muircheartaigh and Campanelli 1999), the intra-interviewer correlation in response indicators is much higher than speculated by Platek and Gray. We deduce then that the contributions of correlated measurement error and nonresponse error variance to overall nonsampling variance will be larger than suggested by Platek and Gray.

7. These intra-interviewer correlations were estimated using PROC GLIMMIX (fitting a one-way random effects logit model to the response indicators) and PROC MIXED (fitting a one-way random effects model to the continuous response deviations) in SAS (Version 9.2). Detailed code for these computations is available from the authors upon request.

Discussion

This study has shown that “large” estimates of ρ_{int} previously observed in studies of interviewer variance may result from significant nonresponse error variance across interviewers in addition to measurement error variance. We used a unique survey data set with record values available for respondents and nonrespondents to conduct a descriptive examination of the relative contributions of nonresponse error variance and measurement error variance. We focused on variables presenting evidence of significant interviewer variance in respondent reports, despite successful interpenetrated assignment of cases to interviewers within calling shifts. We found evidence that interviewer-related variance on some key survey items may be due to nonresponse error variance, that is, differences in respondent characteristics across interviewers, rather than measurement difficulties. We also found that measurement error variance is the primary source of interviewer variance for some estimates.

One of the survey items on which we found significant interviewer variance—the age of the respondent at divorce—was created from the sampled person’s divorce date and birth date. This suggests that different interviewers may successfully recruit respondents of different ages, despite successful interpenetration of sample pools based on age. When analyzing interviewer variance in the ages of the respondents according to the divorce records, there was weak evidence (estimated variance component = 1.68, RLRT statistic = 0.59, $p = 0.19$) of variance among second-shift interviewers in the mean ages of respondents. Notably, the shift of interview was taken into account during these analyses, so the results are not due to different interviewers working different hours. Interviewers may be more successful at recruiting respondents around the same age as themselves, consistent with liking theory (Durrant et al. 2010). Alternatively, certain interviewers may be more proficient with particular age groups (e.g., talking slowly for the elderly), regardless of the interviewers’ age. Why different interviewers recruited respondents of different ages is not knowable from these data, unfortunately; interviewer characteristics such as age or experience level were not available. In general, more work is needed to assess whether certain types of survey items are more or less susceptible to nonresponse error variance or measurement error variance among interviewers.

This study considered one method of assigning nonrespondents to CATI interviewers, which is a difficult problem in general. Importantly, there was little evidence of differential assignment of cases to interviewers *prior* to the survey interviews; only after the interview occurred did notable differences across interviewers occur in the survey reports. In addition, the study results did not change when the interviewer was randomly assigned from all of the interviewers who had ever called a case that had not been contacted. The sensitivity of these results to alternative methods should be examined in future research. The findings from this study would also be strengthened by replication. In an ideal study, a large number of interviewers would be assigned random

subsamples of respondents, only one interviewer would work cases, and validation data would be available for respondents and nonrespondents. This design, although expensive and perhaps difficult to manage, would greatly facilitate our understanding of this phenomenon. Replication of this study using a larger sample of interviewers would also increase the power of analyses to detect significant interviewer variance components.

Interviewer training efforts are often directed at standardizing the administration of survey questions (e.g., Mangione, Fowler, and Louis 1992). This study suggests that interviewer training efforts should also emphasize minimizing differences between respondents and nonrespondents across interviewers. That is, interviewers and survey managers should also focus their efforts on decreasing differential nonresponse error across interviewers, rather than purely trying to decrease nonresponse rates (and measurement errors). Survey managers might consider systems that compare frame information for each interviewer's respondents with frame information for each interviewer's full sample. Supervisors could then intervene to minimize these differences across interviewers.

A limitation of any study of nonresponse and measurement error is that nonrespondents do not provide reported values. The descriptive approach that we employed assumed that systematic measurement errors for a given interviewer would also apply to nonrespondents had they been successfully recruited. This assumption may not be reasonable, given that individuals with a lower propensity to cooperate may be harder to measure on certain survey items (Olson 2006). A well-specified imputation model could impute measurement errors for nonrespondents, and a new dependent variable could be computed for *all* sample units assigned to each interviewer, representing the true value of a survey variable *plus* the measurement error, as previously suggested by Biemer (1980) and Platek and Gray (1983). A multilevel model with sampled units nested within interviewers could then be fit, predicting this new dependent variable with a random intercept and an indicator of survey participation, with a randomly varying coefficient, to disentangle these error sources. The relative variances and covariance of these two random interviewer effects could be examined using this multilevel modeling approach.

The main purpose of this study was to empirically assess whether interviewer variance may be driven by nonresponse error variance among interviewers. We hope that these findings motivate additional analytic development of more appropriate estimators of ρ_{int} in surveys with less than 100-percent response rates. Although this study focused on overall nonresponse outcomes, interviewers may differ in their ability to contact and secure cooperation from sampled units. Groves and Magilavy (1984) present a derivation of the mean squared error (MSE) of an unadjusted respondent mean accounting for interviewers, measurement error, refusal error, and noncontact error. Their approach could spur future derivations of estimators of intra-interviewer correlations for sample means of refusals and noncontacts. These intra-interviewer correlations could then

be used to compute multiplicative interviewer effects on more specific variance components of the MSE, indicating where resources should be targeted for minimizing effects of interviewer variance. Future methodological studies of interviewer variance should continue to consider multiple error sources that interviewers might affect based on a total survey error framework.

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